

SYSTEMS PROJECT MANAGEMENT

Systems Analysis & Design

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FRANCISCO JOSÉ DE CALDAS

Outline

- systems*
- 1 Information Systems / *Data → Info*
 - 2 Enterprises as Systems / *Business*
 - 3 Software Methodologies / *software as a solution*



Outline

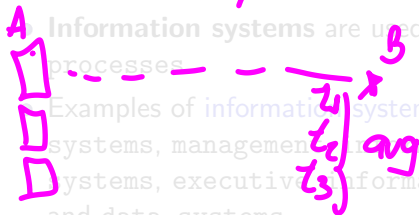
- 1 Information Systems
- 2 Enterprises as Systems
- 3 Software Methodologies



Information Systems

- An **Information System** is a **system** that *collects, processes, stores, and disseminates information.*

↳ Dashboard



Google maps

①

②

Data Scientist

SQL
Document
Objects
Graphs



Information Systems

- An **Information System** is a **system** that *collects, processes, stores, and disseminates* information.
- **Information systems** are used to **support** and **manage** business operations. *Processes + data*
- **Information systems** are used to **automate** and **optimize** business processes. *steps* *computation*
- Examples of information systems include transaction processing systems, management information systems, decision support systems, executive information systems, expert systems, and data systems.



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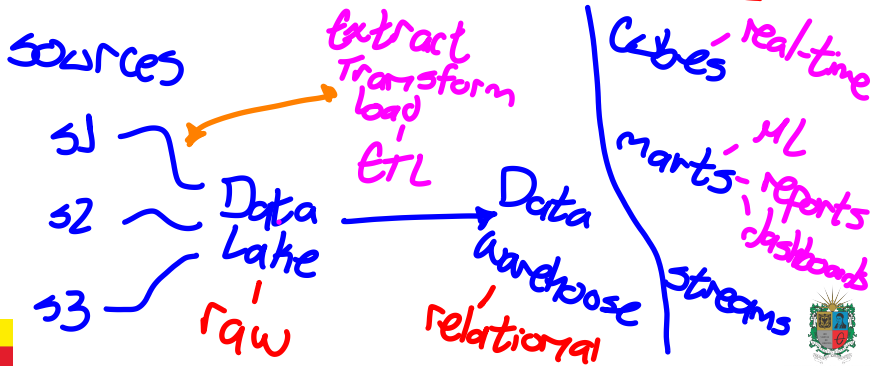
c-level

transversal



Data Systems

- A **Data System** is a system that collects, processes, stores, and retrieves data. *~ real-time*
 → **read / SELECT**
- **Data systems** are used to store and analyze data. **7/24**
- Examples of data systems include databases, data warehouses, data lakes, data marts, data cubes, and data streams.



Expert Systems

- An **Expert System** is a system that uses knowledge and reasoning to solve problems. *Bot*
- **Expert systems** are used to automate and optimize decision-making processes. *~1990s*
- Examples of expert systems include diagnostic systems, predictive systems, prescriptive systems, decision support systems, and automated reasoning systems.



Expert Systems

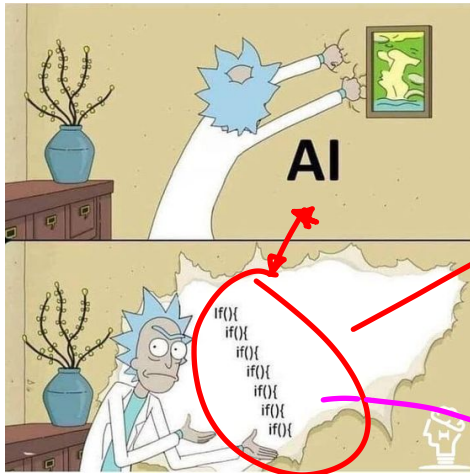
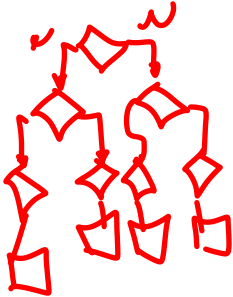
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recommendations



Expert Systems as Classical Artificial Intelligence

Here there is a great example of a **diagnostic system**.



Random forest

decision tree

business



Risks and Failures in Information

- **Information systems** are subject to **risks** and **failures** that can impact business operations.

- **Risks and failures** can be mitigated through **security measures**, **backup systems**, **disaster recovery plans**, and **monitoring tools**.

- Examples of risks and failures include security breaches, data loss, system downtime, performance issues, and compliance violations.

incrementals
today

new

minimize — occurrences
 \ impact

observability
↳ alerts



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memory
time

2-step factor
auth.
store data



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Enterprises: Bottom-Up and Top-Down Approaches

- **Bottom-Up Approach:** Analyzes an enterprise by examining its individual units or components, then aggregating them to understand the entire organization.
- **Top-Down Approach:** Starts with an overall vision or strategy and decomposes it into subsystems, departments, and processes.



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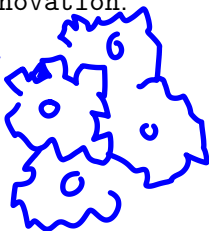
PIECE Framework for Enterprises

- **P**articipation: *Engaging stakeholders* at every level.
- **I**ndependence of Thought: Encouraging *diverse, innovative ideas*.
- **E**laboration: Developing and *refining ideas and processes*.
- **C**ommunication: Ensuring clear, *effective exchange* of information. *→ agile*
- **E**xploration: Embracing *continuous innovation* and *improvement*.



Enterprise System Typologies

- **Rational Systems:** Organizations driven by logical, structured processes and clear hierarchies.
- **Natural Systems:** Organizations viewed as self-organizing entities with emergent behavior. *→ Machine*
- **Open Systems:** Enterprises that continuously interact with their external environment for information, resources, and innovation. *→ Humans - Persons*



*Human
System*



Business Systems and Models

- **Business Systems:** Frameworks that encompass an enterprise's internal processes, operations, and strategies.
- **Examples:** ERP systems, CRM systems, SCM systems.
- **Business Models:** Describe how an organization creates, delivers,

ERP - Enterprise Resource Planning
 CRM - Customer Resource Management
 SCM - Supply Chain Management



Business Systems and Models

- **Business Systems:** Frameworks that encompass an enterprise's internal processes, operations, and strategies.
- **Examples:** ERP systems, CRM systems, SCM systems.
- **Business Models:** Describe how an organization creates, delivers, and captures value.
 - Examples include subscription-based freemium, platform-based, and direct sales models.

Netflix
Podify
cloud



Business Models Examples

Understanding Business Models Through Flywheels

amazon



NETFLIX



Spotify



LinkedIn



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Basic Concepts in Software Methodologies

- **Software methodologies** provide frameworks for **planning**, **designing**, **developing**, **testing**, and **maintaining** software projects.
- They help teams manage project **complexity** and ensure quality deliverables.

system
analysis

quality

legacy

acceptance
close the
project

6 months
3 persons
40 hours/week



Traditional Methodologies

~ 1960-1970

- **Waterfall**: A linear approach where each phase must be *completed before moving* to the next.

Requirements

↳ Analysis & Design

↳ Implementation

↳ Testing

↳ Deployment

Numerical methods

NASA
military

Physical applications



Traditional Methodologies

- **Waterfall:** A **linear approach** where each phase must be *completed before moving* to the next.
- Suitable for projects with **well-defined requirements** and *low uncertainty*.
- Emphasize thorough **documentation** and **planning**.

20-30
artifacts

rigid

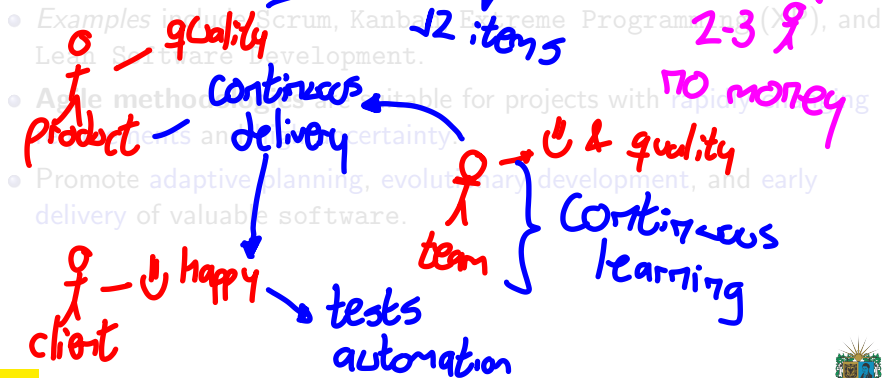
until
80's late



Agile Methodologies

~ 90's internet startups

- Emphasize iterative development, customer collaboration, and flexibility. → **model business**
- Based on the **Agile Manifesto**, which values individuals and interactions over processes and tools.



Agile Methodologies

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- Based on the **Agile Manifesto**, which values individuals and interactions over processes and tools.
- Examples include Scrum ¹⁹⁹⁵ Kanban ^{Management}, Extreme Programming (XP), and Lean Software Development.

Agile methodologies are suitable for projects with rapidly changing requirements and high uncertainty.

~2000's

startup

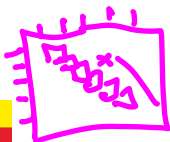
Pair Programming

2 2 2 2 } on-boarding incidents



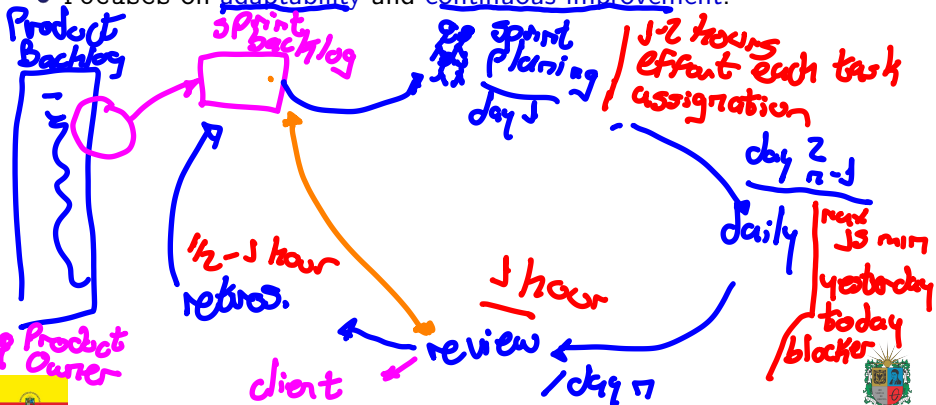
Agile Methodologies

- Emphasize **iterative development**, **customer collaboration**, and **flexibility**.
- Based on the **Agile Manifesto**, which values **individuals** and **interactions** over processes and tools.
- *Examples* include Scrum, Kanban, Extreme Programming (XP), and Lean Software Development.
- **Agile methodologies** are suitable for projects with **rapidly changing requirements** and **high uncertainty**. *→ startup* *Only fans*
- Promote **adaptive planning**, **evolutionary development**, and **early delivery** of valuable software. *2-3 weeks*



Case Study: Scrum

- **Scrum** employs short, time-boxed iterations called sprints.
- Key practices include daily stand-ups, sprint planning, reviews, and retrospectives.
- Focuses on adaptability and continuous improvement.



Case Study: Kanban

- **Kanban** visualizes work items on **boards** and limits Work In Progress (WIP).
- Emphasizes **gradual improvements**, **flow management**, and **continuous delivery**.
- Ideal for **projects** requiring **flexibility** with *minimal iteration planning*.



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

